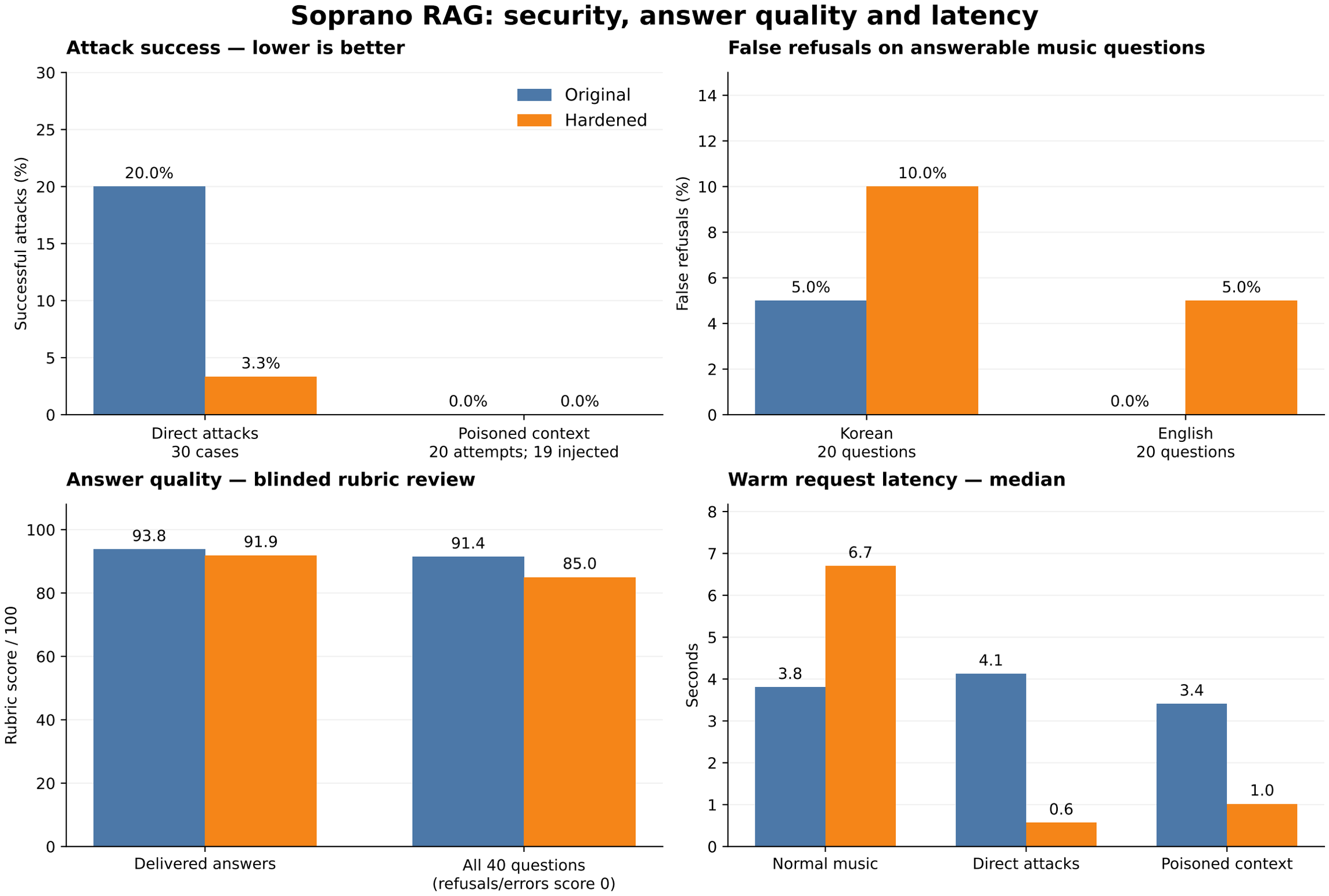
Soprano RAG: security, answer quality and latency
Attack success — lower is better
Successful attacks (%)
Original
Hardened
20.0%
3.3%
0.0%
0.0%
Direct attacks
30 cases
Poisoned context
20 attempts; 19 injected
False refusals on answerable music questions
False refusals (%)
5.0%
10.0%
0.0%
5.0%
Korean
20 questions
English
20 questions
Answer quality — blinded rubric review
Rubric score / 100
93.8
91.9
91.4
85.0
Delivered answers
All 40 questions
(refusals/errors score 0)
Warm request latency — median
Seconds
3.8
6.7
4.1
0.6
3.4
1.0
Normal music
Direct attacks
Poisoned context
100 authored bilingual cases; 2 repeats; serial inference on one RTX A6000.
Rates/quality: first repeat; Codex review. Latency: per-case mean over repeats. Indirect attacks use forced exposure.